CHAPTER 2 · CONCEPT 2 · ANSWER KEY

Answer Key — Cell Structure, Cell Types and Levels of Organisation

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(c) cell wall** — the cell wall is present in plant cells, absent in animal cells.

Q2 **(b) nucleus** — the nucleus regulates activities and growth.

Q3 **(b) a nucleoid (no nuclear membrane)** — bacteria have a nucleoid, not a well-defined nucleus.

Q4 **(b) Cell → Tissue → Organ → Organ system → Organism** — cells build up to tissue, organ, organ system, then organism.

Q5 **(a) Both true, R explains A** — the spindle shape suits its contracting function — shape follows function.

Q6 **safranin** — safranin gives onion cells a pink/red colour.

Q7 **unicellular** — one-celled organisms are unicellular.

Short answer

Q8 Cell membrane — encloses the cell, lets materials in/out; Cytoplasm — where most life processes occur; Nucleus — controls activities and growth.

Q9 Onion cells are rectangular with a cell wall; cheek cells are polygonal with no cell wall (onion stained with safranin, cheek with methylene blue).

Q10 Cell → Tissue → Organ → Organ system → Organism.

Long / application

Q11 **Differences:** plant has cell wall / animal does not; plant has chloroplasts / animal does not; plant has one large vacuole / animal small or none. **Similarities:** both have a cell membrane, cytoplasm and a nucleus.

Q12 Nerve cells are long and branched to carry messages to distant parts quickly; muscle cells are spindle-shaped and flexible to contract and relax (e.g. push food). Each special shape suits the job the cell does.

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